18.085 Computational Science and Engineering I Fall 2026
Recitation 2: two practice papers for Quiz 2

Monday, November 16, 2026

This file holds two practice papers, 2a and 2b, each in the format of Quiz 2: four questions, 20 points each, and on the quiz only your three best answers count. The quiz covers Lectures 10 to 15: Gram–Schmidt and QR, least squares, springs and masses, oscillations and phase space, and the time-stepping methods (forward and backward Euler, leapfrog, trapezoidal, RK4). Nothing on the quiz asks for code or for a proof.

The two papers ask the same four kinds of question with different data, so that the second is a fresh run at whatever went wrong in the first. Try each under quiz conditions: 90 minutes, one letter-size sheet of handwritten notes, a calculator without graphing. The solutions to both papers follow the questions, written out in full with every intermediate number, so that you can find where a calculation of yours went a different way.

Paper 2a, questions	page 2
Paper 2b, questions	page 6
Solutions to paper 2a	page 10
Solutions to paper 2b	page 15

Paper 2a: questions

Question 1 (20 points) Gram–Schmidt and the best line

Five measurements (x_i, y_i) were taken:

x	-2	-1	1	3	7
y	1	4	5	12	19

We want the line $y = kx + b$ that fits them best in the least-squares sense. Write the parameter vector as $\boldsymbol{k} = (k, b)^{\mathsf{T}}$, so that the five equations $kx_i + b = y_i$ read $\boldsymbol{X}\boldsymbol{k} = \boldsymbol{y}$.

(a) *(3 points)* Write down the 5×2 matrix $\boldsymbol{X}$ and the vector $\boldsymbol{y}$. Can all five equations hold at once? (Check with the first two points and the last one.)

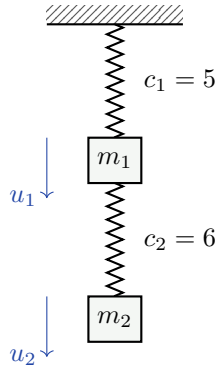
(b) *(7 points)* Apply Gram–Schmidt to the two columns of $\boldsymbol{X}$, in the order given, to find orthonormal vectors $\boldsymbol{q}_1, \boldsymbol{q}_2$ and the upper triangular $\boldsymbol{R}$ with $\boldsymbol{X} = \boldsymbol{Q}\boldsymbol{R}$. (All lengths come out as whole numbers.) Check that $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{q}_2 = 0$.

(c) *(5 points)* Write down the normal equations $\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}\,\boldsymbol{k} = \boldsymbol{X}^{\mathsf{T}}\boldsymbol{y}$ and solve them for k and b.

(d) *(5 points)* Compute $\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{y}$ and solve $\boldsymbol{R}\,\boldsymbol{k} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{y}$ by back-substitution. Do you get the same k and b? Then compute the residual $\bar{\boldsymbol{e}} = \boldsymbol{y} - \boldsymbol{X}\bar{\boldsymbol{k}}$ for the best line, check that it is orthogonal to both columns of $\boldsymbol{X}$, and give $\|\bar{\boldsymbol{e}}\|^2$.

Question 2 (20 points) A chain of two masses

Two masses hang in a chain from a fixed support. The upper spring has stiffness $c_1 = 5$, the lower spring $c_2 = 6$, both masses are $m = 1$, and the lower end is free (units omitted). The displacements u_1, u_2 of the masses are measured downward from the unstretched position, $\boldsymbol{e} = \boldsymbol{A}\boldsymbol{u}$ are the elongations of the springs, $\boldsymbol{w} = \boldsymbol{C}\boldsymbol{e}$ the spring forces, and $\boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}$.



(a) *(6 points)* Write down $\boldsymbol{A}$ and $\boldsymbol{C}$ and compute $\boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}$. Steady forces $\boldsymbol{f} = (4, 6)$ are applied (a pull of 4 on the upper mass and 6 on the lower one). Solve $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}$ for the displacements, and find the elongations $\boldsymbol{e}$ and the spring forces $\boldsymbol{w}$. Check that $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{w} = \boldsymbol{f}$.

(b) *(6 points)* Now the forces are removed and the chain oscillates freely: $\ddot{\boldsymbol{u}} + \boldsymbol{K}\boldsymbol{u} = \boldsymbol{0}$. The general solution is

$$\boldsymbol{u}(t) = (a_1 \cos \omega_1 t + b_1 \sin \omega_1 t)\, \boldsymbol{q}_1 + (a_2 \cos \omega_2 t + b_2 \sin \omega_2 t)\, \boldsymbol{q}_2.$$

Find the two frequencies $\omega_1 < \omega_2$ and the two mode shapes $\boldsymbol{q}_1, \boldsymbol{q}_2$. In one sentence each, describe how the two masses move in each mode.

(c) *(4 points)* The chain is pulled to $\boldsymbol{u}(0) = (5, 1)$ and released from rest. Find $\boldsymbol{u}(t)$.

(d) *(4 points)* With $\boldsymbol{v} = \dot{\boldsymbol{u}}$, write $\ddot{\boldsymbol{u}} + \boldsymbol{K}\boldsymbol{u} = \boldsymbol{0}$ as a first-order system for $\boldsymbol{Y} = (u_1, u_2, v_1, v_2)$. Write the forward Euler step as $\boldsymbol{Y}_{n+1} = \boldsymbol{G}_F\, \boldsymbol{Y}_n$ with its 4×4 matrix $\boldsymbol{G}_F$ (in terms of h and the entries of $\boldsymbol{K}$), and carry out two steps with $h = 0.1$ from the starting values of (c).

Question 3 (20 points) Precision of time steps

The oscillator $\ddot{u} + u = 0$ with $u(0) = 0$, $\dot{u}(0) = 1$ is written as the first-order system $\dot{\boldsymbol{Y}} = \boldsymbol{A}\boldsymbol{Y}$ with

$$\boldsymbol{Y} = \begin{bmatrix} u \\ v \end{bmatrix}, \qquad v = \dot{u}, \qquad \boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

The true solution $(u, v) = (\sin t, \cos t)$ starts at the top of the unit circle of the (u, v) plane, moves clockwise round it, and turns by the angle h (in radians) in each time step h. In this question $h = 0.5$: the true motion turns by 0.5 rad $= 28.65°$ per step and one period is $4\pi \approx 12.57$ steps. Recall the methods ($\boldsymbol{Y}_n$ is the computed value at $t_n = nh$):

forward Euler	$\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + h\boldsymbol{A}\boldsymbol{Y}_n$
backward Euler	$\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + h\boldsymbol{A}\boldsymbol{Y}_{n+1}$
trapezoidal	$\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + \frac{h}{2}\boldsymbol{A}(\boldsymbol{Y}_n + \boldsymbol{Y}_{n+1})$
leapfrog	$V_{1/2} = V_0 - \frac{h}{2}U_0$, then $U_{n+1} = U_n + hV_{n+1/2}$, $V_{n+3/2} = V_{n+1/2} - hU_{n+1}$
RK4	$\boldsymbol{k}_1 = \boldsymbol{A}\boldsymbol{Y}_n$, $\boldsymbol{k}_2 = \boldsymbol{A}(\boldsymbol{Y}_n + \frac{h}{2}\boldsymbol{k}_1)$, $\boldsymbol{k}_3 = \boldsymbol{A}(\boldsymbol{Y}_n + \frac{h}{2}\boldsymbol{k}_2)$, $\boldsymbol{k}_4 = \boldsymbol{A}(\boldsymbol{Y}_n + h\boldsymbol{k}_3)$, $\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + \frac{h}{6}(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4)$

(a) *(5 points)* For $h = 0.5$, write the growth matrices $\boldsymbol{G}_F$, $\boldsymbol{G}_B$ and $\boldsymbol{G}_T$ of forward Euler, backward Euler and the trapezoidal method (so that $\boldsymbol{Y}_{n+1} = \boldsymbol{G}\,\boldsymbol{Y}_n$), and compute $\boldsymbol{Y}_1$ for each. For leapfrog compute $V_{1/2}$, U_1, $V_{3/2}$ and U_2.

(b) *(6 points)* Find the eigenvalues λ of $\boldsymbol{G}_F$, $\boldsymbol{G}_B$, $\boldsymbol{G}_T$ and of the leapfrog matrix

$$\boldsymbol{G}_L = \begin{bmatrix} 1 & h \\ -h & 1 - h^2 \end{bmatrix}, \qquad \text{which maps } (U_n, V_{n+1/2}) \text{ to } (U_{n+1}, V_{n+3/2}).$$

For each method give $|\lambda|$, the factor by which the amplitude changes per step, and the angle θ of λ in degrees, the turn per step. Compare with the true values $|\lambda| = 1$ and $\theta = 28.65°$.

(c) *(5 points)* Over one period of the true motion ($4\pi \approx 12.57$ steps): by what factor do the forward Euler and backward Euler amplitudes grow or shrink? By how many degrees does the trapezoidal method fall behind the true solution, and leapfrog run ahead of it? After how many periods is the trapezoidal solution a quarter of a turn (90°) behind the true one, and after how many is leapfrog a quarter of a turn ahead?

(d) *(4 points)* Carry out one RK4 step with $h = 0.5$ from $\boldsymbol{Y}_0 = (0, 1)$: give $\boldsymbol{k}_1, \ldots, \boldsymbol{k}_4$ and $\boldsymbol{Y}_1$ as exact fractions. Compare with the true point $(\sin 0.5, \cos 0.5) = (0.4794, 0.8776)$ and with the one-step results of (a): which method lands closest?

Question 4 (20 points) Energy in phase space

A mass $m = 1$ kg on a spring with $c = 9$ N/m obeys $\ddot{u} + 9u = 0$, so $\omega = 3$ rad/s. It starts at $u(0) = 1$ m with $v(0) = \dot{u}(0) = 0$. Its energy is

$$E = \tfrac{1}{2}mv^2 + \tfrac{1}{2}cu^2 = \tfrac{1}{2}v^2 + \tfrac{9}{2}u^2.$$

(a) *(4 points)* Write the system as $\dot{\boldsymbol{Y}} = \boldsymbol{A}\boldsymbol{Y}$ with $\boldsymbol{Y} = (u, v)$. What is E at $t = 0$? Sketch the orbit of the true solution in the (u, v) plane: give its equation, mark where it crosses the axes and the direction of motion. What is the period T?

(b) *(6 points)* Forward Euler with $h = 0.1$: compute $\boldsymbol{Y}_1$ and $\boldsymbol{Y}_2$ and the energy after each step. By what factor does E change per step? How many steps does it take for E to double, and by what factor has E grown after one period?

(c) *(5 points)* Backward Euler with $h = 0.1$: compute $\boldsymbol{Y}_1$ and its energy (the growth matrix is $\boldsymbol{G}_B = (\boldsymbol{I} - h\boldsymbol{A})^{-1}$). Then the trapezoidal method with the larger step $h = \frac{1}{3}$: compute $\boldsymbol{G}_T = (\boldsymbol{I} - \frac{h}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{h}{2}\boldsymbol{A})$ and $\boldsymbol{Y}_1$, and check whether $\boldsymbol{Y}_1$ lies on the orbit of (a).

(d) *(5 points)* Leapfrog with $h = 0.1$, for this oscillator:

$$V_{1/2} = V_0 - \tfrac{h}{2}\,\omega^2 U_0, \qquad U_{n+1} = U_n + hV_{n+1/2}, \qquad V_{n+3/2} = V_{n+1/2} - h\,\omega^2 U_{n+1}.$$

Compute $V_{1/2}$, U_1, $V_{3/2}$ and U_2, and check that $Q_n = \omega^2 U_n U_{n+1} + V_{n+1/2}^2$ has the same value for $n = 0$ and $n = 1$. For which steps h does leapfrog stay stable for this oscillator, and how many steps per period is that at least? Finally, say in a few words how E behaves over many periods for each of the four methods.

Paper 2b: questions

Question 1 (20 points) Gram–Schmidt and the best line

Five measurements (x_i, y_i) were taken:

x	-4	0	2	5	6
y	19	9	5	0	-1

We want the line $y = kx + b$ that fits them best in the least-squares sense. Write the parameter vector as $\boldsymbol{k} = (k, b)^{\mathsf{T}}$, so that the five equations $kx_i + b = y_i$ read $\boldsymbol{X}\boldsymbol{k} = \boldsymbol{y}$.

(a) *(3 points)* Write down the 5×2 matrix $\boldsymbol{X}$ and the vector $\boldsymbol{y}$. Can all five equations hold at once? (Check with the first two points and the last one.)

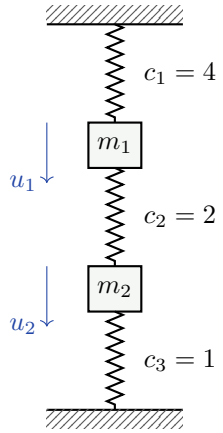
(b) *(7 points)* Apply Gram–Schmidt to the two columns of $\boldsymbol{X}$, in the order given, to find orthonormal vectors $\boldsymbol{q}_1, \boldsymbol{q}_2$ and the upper triangular $\boldsymbol{R}$ with $\boldsymbol{X} = \boldsymbol{Q}\boldsymbol{R}$. (All lengths come out as whole numbers.) Check that $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{q}_2 = 0$.

(c) *(5 points)* Write down the normal equations $\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}\,\boldsymbol{k} = \boldsymbol{X}^{\mathsf{T}}\boldsymbol{y}$ and solve them for k and b.

(d) *(5 points)* Compute $\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{y}$ and solve $\boldsymbol{R}\,\boldsymbol{k} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{y}$ by back-substitution. Do you get the same k and b? Then compute the residual $\bar{\boldsymbol{e}} = \boldsymbol{y} - \boldsymbol{X}\bar{\boldsymbol{k}}$ for the best line, check that it is orthogonal to both columns of $\boldsymbol{X}$, and give $\|\bar{\boldsymbol{e}}\|^2$.

Question 2 (20 points) A chain of two masses

Two masses are held between a ceiling and a floor by three springs, of stiffnesses $c_1 = 4$, $c_2 = 2$ and $c_3 = 1$ from the top down. Both masses are $m = 1$ (units omitted). The displacements u_1, u_2 of the masses are measured downward from the unstretched position, $\boldsymbol{e} = \boldsymbol{A}\boldsymbol{u}$ are the elongations of the three springs, $\boldsymbol{w} = \boldsymbol{C}\boldsymbol{e}$ the spring forces, and $\boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}$.



(a) *(6 points)* Write down the 3×2 matrix $\boldsymbol{A}$ and the 3×3 matrix $\boldsymbol{C}$, and compute $\boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}$. Steady forces $\boldsymbol{f} = (6, 5)$ are applied (a pull of 6 on the upper mass and 5 on the lower one). Solve $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}$ for the displacements, and find the elongations $\boldsymbol{e}$ and the spring forces $\boldsymbol{w}$. Check that $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{w} = \boldsymbol{f}$. Which spring is compressed?

(b) *(6 points)* Now the forces are removed and the chain oscillates freely: $\ddot{\boldsymbol{u}} + \boldsymbol{K}\boldsymbol{u} = \boldsymbol{0}$. The general solution is

$$\boldsymbol{u}(t) = (a_1 \cos \omega_1 t + b_1 \sin \omega_1 t)\, \boldsymbol{q}_1 + (a_2 \cos \omega_2 t + b_2 \sin \omega_2 t)\, \boldsymbol{q}_2.$$

Find the two frequencies $\omega_1 < \omega_2$ and the two mode shapes $\boldsymbol{q}_1, \boldsymbol{q}_2$. In one sentence each, describe how the two masses move in each mode.

(c) *(4 points)* The chain is pulled to $\boldsymbol{u}(0) = (3, 1)$ and released from rest. Find $\boldsymbol{u}(t)$.

(d) *(4 points)* With $\boldsymbol{v} = \dot{\boldsymbol{u}}$, write $\ddot{\boldsymbol{u}} + \boldsymbol{K}\boldsymbol{u} = \boldsymbol{0}$ as a first-order system for $\boldsymbol{Y} = (u_1, u_2, v_1, v_2)$. Write the forward Euler step as $\boldsymbol{Y}_{n+1} = \boldsymbol{G}_F\, \boldsymbol{Y}_n$ with its 4×4 matrix $\boldsymbol{G}_F$ (in terms of h and the entries of $\boldsymbol{K}$), and carry out two steps with $h = 0.1$ from the starting values of (c).

Question 3 (20 points) Precision of time steps

The oscillator $\ddot{u} + 4u = 0$ (so $\omega = 2$) with $u(0) = 1$, $\dot{u}(0) = 0$ is written as the first-order system $\dot{\boldsymbol{Y}} = \boldsymbol{A}\boldsymbol{Y}$ with

$$\boldsymbol{Y} = \begin{bmatrix} u \\ v \end{bmatrix}, \qquad v = \dot{u}, \qquad \boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -4 & 0 \end{bmatrix}.$$

The true solution $(u, v) = (\cos 2t, -2\sin 2t)$ moves clockwise round the ellipse $u^2 + v^2/4 = 1$ of the (u, v) plane. In the stretched coordinates $(u, v/2)$ it is a circle, and the point turns by the angle ωh (in radians) in each time step h; the exact step multiplies $\boldsymbol{Y}$ by a matrix with eigenvalues $e^{\pm i\omega h}$. In this question the step is deliberately coarse, $h = 0.5$: the true motion turns by $\omega h = 1$ rad $= 57.3°$ per step and one period is $\pi/0.5 = 2\pi \approx 6.28$ steps. Recall the methods ($\boldsymbol{Y}_n$ is the computed value at $t_n = nh$):

forward Euler	$\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + h\boldsymbol{A}\boldsymbol{Y}_n$
backward Euler	$\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + h\boldsymbol{A}\boldsymbol{Y}_{n+1}$
trapezoidal	$\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + \frac{h}{2}\boldsymbol{A}(\boldsymbol{Y}_n + \boldsymbol{Y}_{n+1})$
leapfrog	$V_{1/2} = V_0 - \frac{h}{2}\omega^2 U_0$, then $U_{n+1} = U_n + hV_{n+1/2}$, $V_{n+3/2} = V_{n+1/2} - h\omega^2 U_{n+1}$
RK4	$\boldsymbol{k}_1 = \boldsymbol{A}\boldsymbol{Y}_n$, $\boldsymbol{k}_2 = \boldsymbol{A}(\boldsymbol{Y}_n + \frac{h}{2}\boldsymbol{k}_1)$, $\boldsymbol{k}_3 = \boldsymbol{A}(\boldsymbol{Y}_n + \frac{h}{2}\boldsymbol{k}_2)$, $\boldsymbol{k}_4 = \boldsymbol{A}(\boldsymbol{Y}_n + h\boldsymbol{k}_3)$, $\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + \frac{h}{6}(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4)$

(a) *(5 points)* For $h = 0.5$, write the growth matrices $\boldsymbol{G}_F$, $\boldsymbol{G}_B$ and $\boldsymbol{G}_T$ of forward Euler, backward Euler and the trapezoidal method (so that $\boldsymbol{Y}_{n+1} = \boldsymbol{G}\,\boldsymbol{Y}_n$), and compute $\boldsymbol{Y}_1$ for each. For leapfrog compute $V_{1/2}$, U_1, $V_{3/2}$ and U_2.

(b) *(6 points)* Find the eigenvalues λ of $\boldsymbol{G}_F$, $\boldsymbol{G}_B$, $\boldsymbol{G}_T$ and of the leapfrog matrix

$$\boldsymbol{G}_L = \begin{bmatrix} 1 & h \\ -\omega^2 h & 1 - \omega^2 h^2 \end{bmatrix}, \qquad \text{which maps } (U_n, V_{n+1/2}) \text{ to } (U_{n+1}, V_{n+3/2}).$$

For each method give $|\lambda|$, the factor by which the amplitude changes per step, and the angle θ of λ in degrees, the turn per step. Compare with the true values $|\lambda| = 1$ and $\theta = 57.3°$.

(c) *(5 points)* Over one period of the true motion ($2\pi \approx 6.28$ steps): by what factor do the forward Euler and backward Euler amplitudes grow or shrink? By how many degrees does the trapezoidal method fall behind the true solution, and leapfrog run ahead of it? After how many steps is the leapfrog solution back exactly at its starting values $(U_0, V_{1/2})$, and how far has the true solution turned by then?

(d) *(4 points)* Carry out one RK4 step with $h = 0.5$ from $\boldsymbol{Y}_0 = (1, 0)$: give $\boldsymbol{k}_1, \ldots, \boldsymbol{k}_4$ and $\boldsymbol{Y}_1$ as exact fractions. Compare with the true point $(\cos 1, -2\sin 1) = (0.5403, -1.6829)$ and with the one-step results of (a): which method lands closest?

Question 4 (20 points) Energy in phase space

A mass $m = 2$ kg on a spring with $c = 2$ N/m obeys $2\ddot{u} + 2u = 0$, that is $\ddot{u} + u = 0$, so $\omega = 1$ rad/s. It starts at $u(0) = 1$ m with $v(0) = \dot{u}(0) = 0$. Its energy is

$$E = \tfrac{1}{2}mv^2 + \tfrac{1}{2}cu^2 = v^2 + u^2.$$

(a) *(4 points)* Write the system as $\dot{\boldsymbol{Y}} = \boldsymbol{A}\boldsymbol{Y}$ with $\boldsymbol{Y} = (u, v)$. What is E at $t = 0$? Sketch the orbit of the true solution in the (u, v) plane: give its equation, mark where it crosses the axes and the direction of motion. What is the period T?

(b) *(6 points)* Forward Euler with $h = 0.25$: compute $\boldsymbol{Y}_1$ and $\boldsymbol{Y}_2$ and the energy after each step. By what factor does E change per step? How many steps does it take for E to double, and by what factor has E grown after one period?

(c) *(5 points)* Backward Euler with $h = 0.25$: compute $\boldsymbol{Y}_1$ and its energy (the growth matrix is $\boldsymbol{G}_B = (\boldsymbol{I} - h\boldsymbol{A})^{-1}$). Then the trapezoidal method with the larger step $h = 0.5$: compute $\boldsymbol{G}_T = (\boldsymbol{I} - \frac{h}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{h}{2}\boldsymbol{A})$ and $\boldsymbol{Y}_1$, and check whether $\boldsymbol{Y}_1$ lies on the orbit of (a).

(d) *(5 points)* Leapfrog with $h = 0.2$, for this oscillator:

$$V_{1/2} = V_0 - \tfrac{h}{2}\,U_0, \qquad U_{n+1} = U_n + hV_{n+1/2}, \qquad V_{n+3/2} = V_{n+1/2} - h\,U_{n+1}.$$

Compute $V_{1/2}$, U_1, $V_{3/2}$ and U_2, and check that $Q_n = U_nU_{n+1} + V_{n+1/2}^2$ has the same value for $n = 0$ and $n = 1$. For which steps h does leapfrog stay stable for this oscillator, and how many steps per period is that at least? Finally, say in a few words how E behaves over many periods for each of the four methods.

Solutions to paper 2a

Question 1 Gram–Schmidt and the best line

(a)

$$\boldsymbol{X} = \begin{bmatrix} -2 & 1 \\ -1 & 1 \\ 1 & 1 \\ 3 & 1 \\ 7 & 1 \end{bmatrix}, \qquad \boldsymbol{y} = \begin{bmatrix} 1 \\ 4 \\ 5 \\ 12 \\ 19 \end{bmatrix}.$$

The first two points give $-2k + b = 1$ and $-k + b = 4$, so $k = 3$ and $b = 7$. Then the last point would need $7k + b = 28$, but $y_5 = 19$. No line passes through all five points, and $\boldsymbol{Xk} = \boldsymbol{y}$ has no solution.

(b) First column $\boldsymbol{a}_1 = (-2, -1, 1, 3, 7)$: $\|\boldsymbol{a}_1\|^2 = 4 + 1 + 1 + 9 + 49 = 64$, so

$$r_{11} = 8, \qquad \boldsymbol{q}_1 = \tfrac{1}{8}\,(-2, -1, 1, 3, 7).$$

Second column $\boldsymbol{a}_2 = (1, 1, 1, 1, 1)$: its component along $\boldsymbol{q}_1$ is $r_{12} = \boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2 = \frac{1}{8}(-2-1+1+3+7) = 1$. Subtract it,

$$\boldsymbol{a}_2' = \boldsymbol{a}_2 - r_{12}\boldsymbol{q}_1 = (1, 1, 1, 1, 1) - \tfrac{1}{8}(-2, -1, 1, 3, 7) = \tfrac{1}{8}\,(10, 9, 7, 5, 1),$$

with $\|\boldsymbol{a}_2'\|^2 = \frac{1}{64}(100 + 81 + 49 + 25 + 1) = \frac{256}{64} = 4$. So $r_{22} = 2$ and $\boldsymbol{q}_2 = \frac{1}{16}\,(10, 9, 7, 5, 1)$:

$$\boldsymbol{Q} = \begin{bmatrix} -1/4 & 10/16 \\ -1/8 & 9/16 \\ 1/8 & 7/16 \\ 3/8 & 5/16 \\ 7/8 & 1/16 \end{bmatrix}, \qquad \boldsymbol{R} = \begin{bmatrix} 8 & 1 \\ 0 & 2 \end{bmatrix}.$$

Check: $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{q}_2 = \frac{1}{128}(-20 - 9 + 7 + 15 + 7) = 0$. Also $\boldsymbol{QR}$ has the columns $8\boldsymbol{q}_1 = \boldsymbol{a}_1$ and $\boldsymbol{q}_1 + 2\boldsymbol{q}_2 = \frac{1}{8}(-2, -1, 1, 3, 7) + \frac{1}{8}(10, 9, 7, 5, 1) = (1, 1, 1, 1, 1) = \boldsymbol{a}_2$.

(c) With $\sum x_i^2 = 64$, $\sum x_i = 8$, $\sum x_i y_i = -2 - 4 + 5 + 36 + 133 = 168$ and $\sum y_i = 41$,

$$\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X} = \begin{bmatrix} 64 & 8 \\ 8 & 5 \end{bmatrix}, \qquad \boldsymbol{X}^{\mathsf{T}}\boldsymbol{y} = \begin{bmatrix} 168 \\ 41 \end{bmatrix}, \qquad \begin{aligned} 64k + 8b &= 168 \\ 8k + 5b &= 41. \end{aligned}$$

Divide the first equation by 8: $8k + b = 21$. Subtract it from the second: $4b = 20$, so $b = 5$, and then $8k = 16$, $k = 2$. The best line is $y = 2x + 5$.

(d)

$$\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{y} = \begin{bmatrix} \boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{y} \\ \boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{y} \end{bmatrix} = \begin{bmatrix} \frac{1}{8}(-2 - 4 + 5 + 36 + 133) \\ \frac{1}{16}(10 + 36 + 35 + 60 + 19) \end{bmatrix} = \begin{bmatrix} 21 \\ 10 \end{bmatrix}.$$

Back-substitution in $\boldsymbol{Rk} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{y}$: $2b = 10$ gives $b = 5$, then $8k + b = 21$ gives $k = 2$. The same line. The best line predicts $\boldsymbol{X}\bar{\boldsymbol{k}} = (1, 3, 7, 11, 19)$, so

$$\bar{\boldsymbol{e}} = \boldsymbol{y} - \boldsymbol{X}\bar{\boldsymbol{k}} = (0, 1, -2, 1, 0), \qquad \boldsymbol{a}_1^{\mathsf{T}}\bar{\boldsymbol{e}} = 0 - 1 - 2 + 3 + 0 = 0, \qquad \boldsymbol{a}_2^{\mathsf{T}}\bar{\boldsymbol{e}} = 0,$$

and $\|\bar{\boldsymbol{e}}\|^2 = 6$. (As a check, $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{y} - \bar{\boldsymbol{k}}^{\mathsf{T}}\boldsymbol{X}^{\mathsf{T}}\boldsymbol{y} = 547 - (336 + 205) = 6$.)

Question 2 A chain of two masses

(a) Spring 1 stretches by u_1, spring 2 by $u_2 - u_1$:

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \qquad \boldsymbol{C} = \begin{bmatrix} 5 & 0 \\ 0 & 6 \end{bmatrix}, \qquad \boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ -6 & 6 \end{bmatrix} = \begin{bmatrix} 11 & -6 \\ -6 & 6 \end{bmatrix}.$$

$\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}$ reads $11u_1 - 6u_2 = 4$ and $-6u_1 + 6u_2 = 6$. Adding the two, $5u_1 = 10$, so $u_1 = 2$ and $u_2 = u_1 + 1 = 3$. Then

$$\boldsymbol{e} = \boldsymbol{A}\boldsymbol{u} = (2, 1), \qquad \boldsymbol{w} = \boldsymbol{C}\boldsymbol{e} = (10, 6),$$

and $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{w} = (10 - 6, 6) = (4, 6) = \boldsymbol{f}$. The upper spring carries both pulls, $4 + 6 = 10$; the lower spring carries only the 6 below it.

(b) $\det(\boldsymbol{K} - \lambda\boldsymbol{I}) = (11 - \lambda)(6 - \lambda) - 36 = \lambda^2 - 17\lambda + 30 = (\lambda - 2)(\lambda - 15)$, so $\lambda_1 = 2$, $\lambda_2 = 15$ and

$$\omega_1 = \sqrt{2} \approx 1.414, \qquad \omega_2 = \sqrt{15} \approx 3.873.$$

For $\lambda = 2$: $\boldsymbol{K} - 2\boldsymbol{I} = \left[\begin{smallmatrix} 9 & -6 \\ -6 & 4 \end{smallmatrix}\right]$ gives $9q_1 = 6q_2$, so $\boldsymbol{q}_1 = (2, 3)$. For $\lambda = 15$: $\boldsymbol{K} - 15\boldsymbol{I} = \left[\begin{smallmatrix} -4 & -6 \\ -6 & -9 \end{smallmatrix}\right]$ gives $4q_1 = -6q_2$, so $\boldsymbol{q}_2 = (3, -2)$. They are orthogonal, as the eigenvectors of a symmetric matrix must be.

Mode 1 ($\omega_1 = \sqrt{2}$): the two masses move down and up together, the lower one one and a half times as far. Mode 2 ($\omega_2 = \sqrt{15}$): the masses move in opposite directions, the upper one one and a half times as far; the stiff lower spring is stretched hard, which is why this mode is much faster.

(c) Released from rest, so $b_1 = b_2 = 0$: at $t = 0$ the cosine terms have zero velocity and the sine terms have velocity $b_1\omega_1\boldsymbol{q}_1 + b_2\omega_2\boldsymbol{q}_2$, which vanishes only if both b's do. The start $(5, 1) = a_1(2, 3) + a_2(3, -2)$ gives $2a_1 + 3a_2 = 5$ and $3a_1 - 2a_2 = 1$, so $a_1 = a_2 = 1$:

$$\boldsymbol{u}(t) = \cos(\sqrt{2}\,t)\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \cos(\sqrt{15}\,t)\begin{bmatrix} 3 \\ -2 \end{bmatrix}, \qquad \begin{aligned} u_1(t) &= 2\cos\sqrt{2}\,t + 3\cos\sqrt{15}\,t, \\ u_2(t) &= 3\cos\sqrt{2}\,t - 2\cos\sqrt{15}\,t. \end{aligned}$$

(d) $\dot{\boldsymbol{u}} = \boldsymbol{v}$ and $\dot{\boldsymbol{v}} = -\boldsymbol{K}\boldsymbol{u}$, that is

$$\dot{\boldsymbol{Y}} = \begin{bmatrix} \boldsymbol{0} & \boldsymbol{I} \\ -\boldsymbol{K} & \boldsymbol{0} \end{bmatrix}\boldsymbol{Y}.$$

Forward Euler takes the right-hand side at the old step: $\boldsymbol{u}_{n+1} = \boldsymbol{u}_n + h\boldsymbol{v}_n$, $\boldsymbol{v}_{n+1} = \boldsymbol{v}_n - h\boldsymbol{K}\boldsymbol{u}_n$, so

$$\boldsymbol{G}_F = \begin{bmatrix} \boldsymbol{I} & h\boldsymbol{I} \\ -h\boldsymbol{K} & \boldsymbol{I} \end{bmatrix} = \begin{bmatrix} 1 & 0 & h & 0 \\ 0 & 1 & 0 & h \\ -11h & 6h & 1 & 0 \\ 6h & -6h & 0 & 1 \end{bmatrix}.$$

From $\boldsymbol{Y}_0 = (5, 1, 0, 0)$ with $h = 0.1$: $\boldsymbol{K}\boldsymbol{u}_0 = (55 - 6, -30 + 6) = (49, -24)$, so

$$\boldsymbol{Y}_1 = (5,\ 1,\ -4.9,\ 2.4),$$

and, using $\boldsymbol{K}\boldsymbol{u}_1 = \boldsymbol{K}\boldsymbol{u}_0$ since $\boldsymbol{u}_1 = \boldsymbol{u}_0$,

$$\boldsymbol{Y}_2 = (5 - 0.49,\ 1 + 0.24,\ -4.9 - 4.9,\ 2.4 + 2.4) = (4.51,\ 1.24,\ -9.8,\ 4.8).$$

(The exact position at $t = 0.2$ from (c) is $(4.065, 1.451)$: forward Euler's position starts moving one step late, as always.)

Question 3 Precision of time steps

(a) With $h = 0.5$:

$$\boldsymbol{G}_F = \boldsymbol{I} + \tfrac{1}{2}\boldsymbol{A} = \begin{bmatrix} 1 & \frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix}, \qquad \boldsymbol{G}_B = (\boldsymbol{I} - \tfrac{1}{2}\boldsymbol{A})^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}^{-1} = \frac{1}{5/4}\begin{bmatrix} 1 & \frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix} = \begin{bmatrix} 0.8 & 0.4 \\ -0.4 & 0.8 \end{bmatrix},$$

and, since $\boldsymbol{I} - \frac{1}{4}\boldsymbol{A} = \left[\begin{smallmatrix} 1 & -\frac{1}{4} \\ \frac{1}{4} & 1 \end{smallmatrix}\right]$ has determinant $\frac{17}{16}$ and inverse $\frac{16}{17}(\boldsymbol{I} + \frac{1}{4}\boldsymbol{A})$,

$$\boldsymbol{G}_T = (\boldsymbol{I} - \tfrac{1}{4}\boldsymbol{A})^{-1}(\boldsymbol{I} + \tfrac{1}{4}\boldsymbol{A}) = \frac{16}{17}\begin{bmatrix} 1 & \frac{1}{4} \\ -\frac{1}{4} & 1 \end{bmatrix}^2 = \frac{16}{17}\begin{bmatrix} \frac{15}{16} & \frac{1}{2} \\ -\frac{1}{2} & \frac{15}{16} \end{bmatrix} = \frac{1}{17}\begin{bmatrix} 15 & 8 \\ -8 & 15 \end{bmatrix}.$$

Applied to $\boldsymbol{Y}_0 = (0, 1)$ they give the second columns:

$$\boldsymbol{Y}_1^F = (0.5, 1), \qquad \boldsymbol{Y}_1^B = (0.4, 0.8), \qquad \boldsymbol{Y}_1^T = (\tfrac{8}{17}, \tfrac{15}{17}) = (0.4706, 0.8824).$$

Leapfrog: $V_{1/2} = 1 - \frac{1}{4}\cdot 0 = 1$; $U_1 = 0 + 0.5\cdot 1 = 0.5$; $V_{3/2} = 1 - 0.5\cdot 0.5 = 0.75$; $U_2 = 0.5 + 0.5\cdot 0.75 = 0.875$. (True values: $\sin 0.5 = 0.479$, $\cos 0.75 = 0.732$, $\sin 1 = 0.841$.)

(b) The eigenvalues come from $\det(\boldsymbol{G} - \lambda\boldsymbol{I}) = 0$; for the three matrices of the form $\left[\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right]$ they are $a \pm ib$. For leapfrog, $\boldsymbol{G}_L = \left[\begin{smallmatrix} 1 & 0.5 \\ -0.5 & 0.75 \end{smallmatrix}\right]$ and $\lambda^2 - 1.75\lambda + 1 = 0$ gives $\lambda = 0.875 \pm i\sqrt{1 - 0.875^2} = 0.875 \pm 0.4841i$.

method	λ	$\lvert\lambda\rvert$ per step	θ per step
forward Euler	$1 \pm 0.5i$	$\sqrt{1.25} = 1.118$ (grows)	$\arctan 0.5 = 26.57°$ (2.1° behind)
backward Euler	$0.8 \pm 0.4i$	$\sqrt{0.8} = 0.894$ (shrinks)	26.57° (2.1° behind)
trapezoidal	$\frac{1}{17}(15 \pm 8i)$	1	$\arctan \frac{8}{15} = 28.07°$ (0.58° behind)
leapfrog	$0.875 \pm 0.484i$	1	$\arccos 0.875 = 28.96°$ (0.31° ahead)
true	$e^{\pm 0.5i}$	1	28.65°

The two Euler methods change the amplitude by about 11% per step and turn too little. The two second-order methods keep the amplitude exactly and are off only in the angle, by a fraction of a degree per step: trapezoidal behind, leapfrog ahead.

(c) A period is $4\pi \approx 12.57$ steps. Forward Euler multiplies the amplitude by $1.118^{12.57} \approx 4.1$ per period; backward Euler by $0.894^{12.57} \approx 0.25$. The trapezoidal method turns $12.57 \times 28.07° = 352.8°$ per period, so it falls behind by $12.57 \times 0.58° \approx 7.2°$ per period; leapfrog turns $12.57 \times 28.96° = 363.9°$, so it runs ahead by $12.57 \times 0.31° \approx 3.9°$ per period. A quarter turn behind takes the trapezoidal method about $90/7.2 \approx 12$ periods; a quarter turn ahead takes leapfrog about $90/3.9 \approx 23$ periods.

(d) Every slope is $\boldsymbol{A}$ applied to a trial point, $\boldsymbol{A}(u, v) = (v, -u)$:

$$\begin{aligned} &\boldsymbol{k}_1 = \boldsymbol{A}\boldsymbol{Y}_0 = (1, 0), \\ &\boldsymbol{Y}_0 + \tfrac{1}{4}\boldsymbol{k}_1 = (\tfrac{1}{4}, 1), \quad &&\boldsymbol{k}_2 = (1, -\tfrac{1}{4}), \\ &\boldsymbol{Y}_0 + \tfrac{1}{4}\boldsymbol{k}_2 = (\tfrac{1}{4}, \tfrac{15}{16}), \quad &&\boldsymbol{k}_3 = (\tfrac{15}{16}, -\tfrac{1}{4}), \\ &\boldsymbol{Y}_0 + \tfrac{1}{2}\boldsymbol{k}_3 = (\tfrac{15}{32}, \tfrac{7}{8}), \quad &&\boldsymbol{k}_4 = (\tfrac{7}{8}, -\tfrac{15}{32}). \end{aligned}$$

Then $\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4 = (1 + 2 + \frac{15}{8} + \frac{7}{8},\ 0 - \frac{1}{2} - \frac{1}{2} - \frac{15}{32}) = (\frac{23}{4}, -\frac{47}{32})$ and

$$\boldsymbol{Y}_1 = (0,1) + \tfrac{1}{12}(\tfrac{23}{4}, -\tfrac{47}{32}) = \Big(\frac{23}{48}, \frac{337}{384}\Big) = (0.47917, 0.87760).$$

Distances from the true point $(0.47943, 0.87758)$: RK4 0.00026, trapezoidal 0.010, backward Euler 0.111, forward Euler 0.124; leapfrog's $U_1 = 0.5$ is 0.021 from $\sin 0.5$ (its velocity sits at a different time). RK4 lands closest by a wide margin, at the cost of four evaluations per step.

Question 4 Energy in phase space

(a) $\dot{u} = v$, $\dot{v} = -9u$:

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -9 & 0 \end{bmatrix}, \qquad E(0) = 4.5 \text{ J}.$$

The true solution $u = \cos 3t$, $v = -3\sin 3t$ keeps $E = 4.5$, so the orbit is the ellipse

$$\tfrac{1}{2}v^2 + \tfrac{9}{2}u^2 = \tfrac{9}{2}, \qquad \text{that is} \qquad u^2 + \frac{v^2}{9} = 1,$$

through $u = \pm 1$ on the u-axis and $v = \pm 3$ on the v-axis. At $(1,0)$ the velocity $\dot{v} = -9 < 0$ points down, so the motion is clockwise. The period is $T = 2\pi/\omega = 2\pi/3 \approx 2.09$ s.

(b) $\boldsymbol{G}_F = \boldsymbol{I} + h\boldsymbol{A} = \left[\begin{smallmatrix} 1 & 0.1 \\ -0.9 & 1 \end{smallmatrix}\right]$.

$$\begin{aligned} &\boldsymbol{Y}_1 = (1, -0.9), && E_1 = \tfrac{1}{2}(0.81) + \tfrac{9}{2}(1) = 4.905, \\ &\boldsymbol{Y}_2 = (1 - 0.09,\ -0.9 - 0.9) = (0.91, -1.8), && E_2 = \tfrac{1}{2}(3.24) + \tfrac{9}{2}(0.8281) = 5.34645. \end{aligned}$$

$E_1/E_0 = 4.905/4.5 = 1.09$ and $E_2/E_1 = 5.34645/4.905 = 1.09$: the energy grows by the factor $1 + \omega^2 h^2 = 1.09$ at every step. (In general $\frac{1}{2}(v - 9hu)^2 + \frac{9}{2}(u + hv)^2 = (1 + 9h^2)(\frac{1}{2}v^2 + \frac{9}{2}u^2)$, the cross terms $-9huv + 9huv$ cancelling.) It doubles when $1.09^n = 2$, that is $n = \ln 2 / \ln 1.09 = 8.04$, so by the 9th step (at $t = 0.9$ s, less than half a period). One period is $2.094/0.1 = 20.9$ steps, so E grows by $1.09^{20.9} \approx 6.1$ per period (the amplitude by $\sqrt{6.1} \approx 2.5$).

(c) $\boldsymbol{I} - h\boldsymbol{A} = \left[\begin{smallmatrix} 1 & -0.1 \\ 0.9 & 1 \end{smallmatrix}\right]$ has determinant 1.09, so

$$\boldsymbol{G}_B = \frac{1}{1.09}\begin{bmatrix} 1 & 0.1 \\ -0.9 & 1 \end{bmatrix} = \frac{\boldsymbol{G}_F}{1.09}, \qquad \boldsymbol{Y}_1 = \frac{(1, -0.9)}{1.09} = (0.9174, -0.8257),$$

and $E_1 = 4.905/1.09^2 = 4.5/1.09 = 4.128$. Backward Euler divides the energy by 1.09 at every step (down to 0.16 of its value after one period).

Trapezoidal with $h = \frac{1}{3}$: $\frac{h}{2}\boldsymbol{A} = \left[\begin{smallmatrix} 0 & 1/6 \\ -3/2 & 0 \end{smallmatrix}\right]$, and $\boldsymbol{I} - \frac{h}{2}\boldsymbol{A} = \left[\begin{smallmatrix} 1 & -1/6 \\ 3/2 & 1 \end{smallmatrix}\right]$ has determinant $1 + \frac{1}{4} = \frac{5}{4}$, so

$$\boldsymbol{G}_T = \frac{4}{5}\begin{bmatrix} 1 & \frac{1}{6} \\ -\frac{3}{2} & 1 \end{bmatrix}\begin{bmatrix} 1 & \frac{1}{6} \\ -\frac{3}{2} & 1 \end{bmatrix} = \frac{4}{5}\begin{bmatrix} \frac{3}{4} & \frac{1}{3} \\ -3 & \frac{3}{4} \end{bmatrix} = \begin{bmatrix} 0.6 & 4/15 \\ -2.4 & 0.6 \end{bmatrix},$$

and $\boldsymbol{Y}_1 = (0.6, -2.4)$. Check: $u^2 + v^2/9 = 0.36 + 0.64 = 1$, and $E_1 = \frac{1}{2}(5.76) + \frac{9}{2}(0.36) = 2.88 + 1.62 = 4.5$. The point is on the ellipse: the trapezoidal method keeps the energy exactly. (The true point at $t = \frac{1}{3}$ is $(\cos 1, -3\sin 1) = (0.540, -2.524)$, a little further round; the error of the trapezoidal method is in the angle, not in the energy.)

(d) With $\omega^2 = 9$ and $h = 0.1$:

$$V_{1/2} = 0 - 0.05 \cdot 9 \cdot 1 = -0.45, \qquad U_1 = 1 + 0.1(-0.45) = 0.955,$$
$$V_{3/2} = -0.45 - 0.1 \cdot 9 \cdot 0.955 = -1.3095, \quad U_2 = 0.955 + 0.1(-1.3095) = 0.82405.$$

$$Q_0 = 9(1)(0.955) + (0.45)^2 = 8.595 + 0.2025 = 8.7975,$$
$$Q_1 = 9(0.955)(0.82405) + (1.3095)^2 = 7.0827 + 1.7148 = 8.7975.$$

The two agree: Q is the quantity leapfrog conserves exactly, at every step. It is $2E$ with u^2 replaced by the product of two neighbouring positions, and $Q_0 = 2E_0(1 - \omega^2 h^2/4) = 9 \times 0.9775$. Leapfrog is stable for $\omega h < 2$, that is $h < \frac{2}{3}$ s; the period is 2.09 s, so that means more than π, hence at least 4, steps per period.

Over many periods: forward Euler's E grows without bound (times 1.09 per step here); backward Euler's decays to zero (divided by 1.09 per step); leapfrog's stays bounded, moving up and down within a fixed band around E_0 while Q is exactly constant; the trapezoidal method's is exactly constant, $E = 4.5$ at every step and for every h.

Solutions to paper 2b

Question 1 Gram–Schmidt and the best line

(a)

$$\boldsymbol{X} = \begin{bmatrix} -4 & 1 \\ 0 & 1 \\ 2 & 1 \\ 5 & 1 \\ 6 & 1 \end{bmatrix}, \qquad \boldsymbol{y} = \begin{bmatrix} 19 \\ 9 \\ 5 \\ 0 \\ -1 \end{bmatrix}.$$

The first two points give $-4k + b = 19$ and $b = 9$, so $k = -2.5$. Then the last point would need $6k + b = -6$, but $y_5 = -1$. No line passes through all five points, and $\boldsymbol{Xk} = \boldsymbol{y}$ has no solution.

(b) First column $\boldsymbol{a}_1 = (-4, 0, 2, 5, 6)$: $\|\boldsymbol{a}_1\|^2 = 16 + 0 + 4 + 25 + 36 = 81$, so

$$r_{11} = 9, \qquad \boldsymbol{q}_1 = \tfrac{1}{9}\,(-4, 0, 2, 5, 6).$$

Second column $\boldsymbol{a}_2 = (1, 1, 1, 1, 1)$: its component along $\boldsymbol{q}_1$ is $r_{12} = \boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2 = \frac{1}{9}(-4+0+2+5+6) = 1$. Subtract it,

$$\boldsymbol{a}_2' = \boldsymbol{a}_2 - r_{12}\boldsymbol{q}_1 = (1, 1, 1, 1, 1) - \tfrac{1}{9}(-4, 0, 2, 5, 6) = \tfrac{1}{9}\,(13, 9, 7, 4, 3),$$

with $\|\boldsymbol{a}_2'\|^2 = \frac{1}{81}(169 + 81 + 49 + 16 + 9) = \frac{324}{81} = 4$. So $r_{22} = 2$ and $\boldsymbol{q}_2 = \frac{1}{18}\,(13, 9, 7, 4, 3)$:

$$\boldsymbol{Q} = \begin{bmatrix} -4/9 & 13/18 \\ 0 & 9/18 \\ 2/9 & 7/18 \\ 5/9 & 4/18 \\ 6/9 & 3/18 \end{bmatrix}, \qquad \boldsymbol{R} = \begin{bmatrix} 9 & 1 \\ 0 & 2 \end{bmatrix}.$$

Check: $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{q}_2 = \frac{1}{162}(-52 + 0 + 14 + 20 + 18) = 0$. Also $\boldsymbol{QR}$ has the columns $9\boldsymbol{q}_1 = \boldsymbol{a}_1$ and $\boldsymbol{q}_1 + 2\boldsymbol{q}_2 = \frac{1}{9}(-4, 0, 2, 5, 6) + \frac{1}{9}(13, 9, 7, 4, 3) = (1, 1, 1, 1, 1) = \boldsymbol{a}_2$.

(c) With $\sum x_i^2 = 81$, $\sum x_i = 9$, $\sum x_i y_i = -76 + 0 + 10 + 0 - 6 = -72$ and $\sum y_i = 32$,

$$\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X} = \begin{bmatrix} 81 & 9 \\ 9 & 5 \end{bmatrix}, \qquad \boldsymbol{X}^{\mathsf{T}}\boldsymbol{y} = \begin{bmatrix} -72 \\ 32 \end{bmatrix}, \qquad \begin{aligned} 81k + 9b &= -72 \\ 9k + 5b &= 32. \end{aligned}$$

Divide the first equation by 9: $9k + b = -8$. Subtract it from the second: $4b = 40$, so $b = 10$, and then $9k = -18$, $k = -2$. The best line is $y = -2x + 10$.

(d)

$$\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{y} = \begin{bmatrix} \boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{y} \\ \boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{y} \end{bmatrix} = \begin{bmatrix} \frac{1}{9}(-76 + 0 + 10 + 0 - 6) \\ \frac{1}{18}(247 + 81 + 35 + 0 - 3) \end{bmatrix} = \begin{bmatrix} -8 \\ 20 \end{bmatrix}.$$

Back-substitution in $\boldsymbol{Rk} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{y}$: $2b = 20$ gives $b = 10$, then $9k + b = -8$ gives $k = -2$. The same line.

The best line predicts $\boldsymbol{X}\bar{\boldsymbol{k}} = (18, 10, 6, 0, -2)$, so

$$\bar{\boldsymbol{e}} = \boldsymbol{y} - \boldsymbol{X}\bar{\boldsymbol{k}} = (1, -1, -1, 0, 1), \qquad \boldsymbol{a}_1^{\mathsf{T}}\bar{\boldsymbol{e}} = -4 + 0 - 2 + 0 + 6 = 0, \qquad \boldsymbol{a}_2^{\mathsf{T}}\bar{\boldsymbol{e}} = 0,$$

and $\|\bar{\boldsymbol{e}}\|^2 = 4$. (As a check, $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{y} - \bar{\boldsymbol{k}}^{\mathsf{T}}\boldsymbol{X}^{\mathsf{T}}\boldsymbol{y} = 468 - (144 + 320) = 4$.)

Question 2 A chain of two masses

(a) Spring 1 stretches by u_1, spring 2 by $u_2 - u_1$, and spring 3, whose lower end is fixed, by $-u_2$:

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix}, \qquad \boldsymbol{C} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ -2 & 2 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -2 \\ -2 & 3 \end{bmatrix}.$$

(Each diagonal entry is the sum of the two springs at that mass, $4+2$ and $2+1$; the off-diagonal entry is minus the spring between them.) $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}$ reads $6u_1 - 2u_2 = 6$ and $-2u_1 + 3u_2 = 5$. From the first, $u_2 = 3u_1 - 3$; in the second, $-2u_1 + 9u_1 - 9 = 5$, so $u_1 = 2$ and $u_2 = 3$. Then

$$\boldsymbol{e} = \boldsymbol{A}\boldsymbol{u} = (2, 1, -3), \qquad \boldsymbol{w} = \boldsymbol{C}\boldsymbol{e} = (8, 2, -3),$$

and $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{w} = (8-2, 2+3) = (6, 5) = \boldsymbol{f}$. Spring 3 has a negative elongation: it is compressed by 3 and pushes up on the lower mass with a force of 3.

(b) $\det(\boldsymbol{K} - \lambda\boldsymbol{I}) = (6-\lambda)(3-\lambda) - 4 = \lambda^2 - 9\lambda + 14 = (\lambda - 2)(\lambda - 7)$, so $\lambda_1 = 2$, $\lambda_2 = 7$ and

$$\omega_1 = \sqrt{2} \approx 1.414, \qquad \omega_2 = \sqrt{7} \approx 2.646.$$

For $\lambda = 2$: $\boldsymbol{K} - 2\boldsymbol{I} = \left[\begin{smallmatrix} 4 & -2 \\ -2 & 1 \end{smallmatrix}\right]$ gives $4q_1 = 2q_2$, so $\boldsymbol{q}_1 = (1, 2)$. For $\lambda = 7$: $\boldsymbol{K} - 7\boldsymbol{I} = \left[\begin{smallmatrix} -1 & -2 \\ -2 & -4 \end{smallmatrix}\right]$ gives $q_1 = -2q_2$, so $\boldsymbol{q}_2 = (2, -1)$. They are orthogonal, as the eigenvectors of a symmetric matrix must be.

Mode 1 ($\omega_1 = \sqrt{2}$): the two masses move down and up together, the lower one twice as far; the middle spring is stretched only a little. Mode 2 ($\omega_2 = \sqrt{7}$): the masses move in opposite directions, the upper one twice as far, and the middle spring is worked hard, which is why this mode is faster.

(c) Released from rest, so $b_1 = b_2 = 0$: at $t = 0$ the cosine terms have zero velocity and the sine terms have velocity $b_1\omega_1\boldsymbol{q}_1 + b_2\omega_2\boldsymbol{q}_2$, which vanishes only if both b's do. The start $(3, 1) = a_1(1, 2) + a_2(2, -1)$ gives $a_1 + 2a_2 = 3$ and $2a_1 - a_2 = 1$, so $a_1 = a_2 = 1$:

$$\boldsymbol{u}(t) = \cos(\sqrt{2}\,t) \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \cos(\sqrt{7}\,t) \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \qquad \begin{aligned} u_1(t) &= \cos\sqrt{2}\,t + 2\cos\sqrt{7}\,t, \\ u_2(t) &= 2\cos\sqrt{2}\,t - \cos\sqrt{7}\,t. \end{aligned}$$

(d) $\dot{\boldsymbol{u}} = \boldsymbol{v}$ and $\dot{\boldsymbol{v}} = -\boldsymbol{K}\boldsymbol{u}$, that is

$$\dot{\boldsymbol{Y}} = \begin{bmatrix} \boldsymbol{0} & \boldsymbol{I} \\ -\boldsymbol{K} & \boldsymbol{0} \end{bmatrix} \boldsymbol{Y}.$$

Forward Euler takes the right-hand side at the old step: $\boldsymbol{u}_{n+1} = \boldsymbol{u}_n + h\boldsymbol{v}_n$, $\boldsymbol{v}_{n+1} = \boldsymbol{v}_n - h\boldsymbol{K}\boldsymbol{u}_n$, so

$$\boldsymbol{G}_F = \begin{bmatrix} \boldsymbol{I} & h\boldsymbol{I} \\ -h\boldsymbol{K} & \boldsymbol{I} \end{bmatrix} = \begin{bmatrix} 1 & 0 & h & 0 \\ 0 & 1 & 0 & h \\ -6h & 2h & 1 & 0 \\ 2h & -3h & 0 & 1 \end{bmatrix}.$$

From $\boldsymbol{Y}_0 = (3, 1, 0, 0)$ with $h = 0.1$: $\boldsymbol{K}\boldsymbol{u}_0 = (18-2, -6+3) = (16, -3)$, so

$$\boldsymbol{Y}_1 = (3,\ 1,\ -1.6,\ 0.3),$$

and, using $\boldsymbol{K}\boldsymbol{u}_1 = \boldsymbol{K}\boldsymbol{u}_0$ since $\boldsymbol{u}_1 = \boldsymbol{u}_0$,

$$\boldsymbol{Y}_2 = (3 - 0.16,\ 1 + 0.03,\ -1.6 - 1.6,\ 0.3 + 0.3) = (2.84,\ 1.03,\ -3.2,\ 0.6).$$

(The exact position at $t = 0.2$ from (c) is $(2.687, 1.057)$: forward Euler's position starts moving one step late, as always.)

Question 3 Precision of time steps

(a) With $h = 0.5$, $h\boldsymbol{A} = \left[\begin{smallmatrix} 0 & 0.5 \\ -2 & 0 \end{smallmatrix}\right]$:

$$\boldsymbol{G}_F = \boldsymbol{I} + h\boldsymbol{A} = \begin{bmatrix} 1 & 0.5 \\ -2 & 1 \end{bmatrix}, \qquad \boldsymbol{G}_B = (\boldsymbol{I} - h\boldsymbol{A})^{-1} = \begin{bmatrix} 1 & -0.5 \\ 2 & 1 \end{bmatrix}^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 0.5 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.25 \\ -1 & 0.5 \end{bmatrix},$$

and, since $\boldsymbol{I} - \frac{h}{2}\boldsymbol{A} = \left[\begin{smallmatrix} 1 & -0.25 \\ 1 & 1 \end{smallmatrix}\right]$ has determinant $\frac{5}{4}$ and inverse $\frac{4}{5}(\boldsymbol{I} + \frac{h}{2}\boldsymbol{A})$,

$$\boldsymbol{G}_T = (\boldsymbol{I} - \tfrac{h}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \tfrac{h}{2}\boldsymbol{A}) = \frac{4}{5}\begin{bmatrix} 1 & 0.25 \\ -1 & 1 \end{bmatrix}^2 = \frac{4}{5}\begin{bmatrix} 0.75 & 0.5 \\ -2 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.6 & 0.4 \\ -1.6 & 0.6 \end{bmatrix}.$$

Applied to $\boldsymbol{Y}_0 = (1, 0)$ they give the first columns:

$$\boldsymbol{Y}_1^F = (1, -2), \qquad \boldsymbol{Y}_1^B = (0.5, -1), \qquad \boldsymbol{Y}_1^T = (0.6, -1.6).$$

Leapfrog, with $\omega^2 = 4$: $V_{1/2} = 0 - 0.25\cdot 4\cdot 1 = -1$; $U_1 = 1 + 0.5\cdot(-1) = 0.5$; $V_{3/2} = -1 - 0.5\cdot 4\cdot 0.5 = -2$; $U_2 = 0.5 + 0.5\cdot(-2) = -0.5$. (True values: $\cos 1 = 0.540$, $-2\sin 1.5 = -1.995$, $\cos 2 = -0.416$.)

(b) For $\boldsymbol{G}_F$: $\det(\boldsymbol{G}_F - \lambda\boldsymbol{I}) = (1-\lambda)^2 + 1 = 0$ gives $\lambda = 1 \pm i$. $\boldsymbol{G}_B = \frac{1}{2}\boldsymbol{G}_F$ has half these, $\frac{1}{2}(1 \pm i)$. For $\boldsymbol{G}_T$: $(0.6 - \lambda)^2 + 0.64 = 0$ gives $0.6 \pm 0.8i$. For $\boldsymbol{G}_L = \left[\begin{smallmatrix} 1 & 0.5 \\ -2 & 0 \end{smallmatrix}\right]$: $\lambda^2 - \lambda + 1 = 0$ gives $\lambda = \frac{1}{2}(1 \pm i\sqrt{3}) = e^{\pm i\pi/3}$.

method	λ	$\lvert\lambda\rvert$ per step	θ per step
forward Euler	$1 \pm i$	$\sqrt{2} = 1.414$ (grows)	$45°$ (turns too little)
backward Euler	$\frac{1}{2}(1 \pm i)$	$1/\sqrt{2} = 0.707$ (shrinks)	$45°$ (turns too little)
trapezoidal	$0.6 \pm 0.8i$	1	$\arctan\frac{4}{3} = 53.13°$ ($4.2°$ behind)
leapfrog	$\frac{1}{2}(1 \pm i\sqrt{3})$	1	$60°$ ($2.7°$ ahead)
true	$e^{\pm i}$	1	$57.30°$

The two Euler methods get the amplitude wrong by a factor $\sqrt{2}$ per step and turn by $45°$ instead of $57.3°$. The two second-order methods keep the amplitude exactly and are off only in the angle, trapezoidal behind, leapfrog ahead. (Only the product ωh enters the eigenvalues: any oscillator stepped with the same ωh has the same $|\lambda|$ and θ.)

(c) A period is $2\pi \approx 6.28$ steps. Forward Euler multiplies the amplitude by $(\sqrt{2})^{2\pi} = 2^\pi \approx 8.8$ per period; backward Euler by $2^{-\pi} \approx 0.11$. The trapezoidal method turns $2\pi \times 53.13° = 333.8°$ per period, so it falls behind by $2\pi\times(57.30 - 53.13)° \approx 26°$ per period. Leapfrog turns $2\pi\times 60° = 377.0°$, so it runs ahead by $2\pi \times 2.70° \approx 17°$ per period.

Since leapfrog turns by exactly $60°$ per step, it is back at $(U_0, V_{1/2}) = (1, -1)$ after 6 steps: $\lambda^6 = e^{\pm 2\pi i} = 1$, so $\boldsymbol{G}_L^6 = \boldsymbol{I}$. Indeed $U_n = \cos(60°n)$ runs through $1, \frac{1}{2}, -\frac{1}{2}, -1, -\frac{1}{2}, \frac{1}{2}, 1$. The true solution at $t = 3$ has turned $\omega t = 6$ rad $= 343.8°$, which is $16.2°$ short of a full turn: $(\cos 6, -2\sin 6) = (0.960, 0.559)$. The leapfrog period is 6 steps against the true 6.28.

(d) Every slope is $\boldsymbol{A}$ applied to a trial point, $\boldsymbol{A}(u, v) = (v, -4u)$:

$$\begin{aligned} &\boldsymbol{k}_1 = \boldsymbol{A}\boldsymbol{Y}_0 = (0, -4), \\ &\boldsymbol{Y}_0 + \tfrac{1}{4}\boldsymbol{k}_1 = (1, -1), \quad \boldsymbol{k}_2 = (-1, -4), \\ &\boldsymbol{Y}_0 + \tfrac{1}{4}\boldsymbol{k}_2 = (\tfrac{3}{4}, -1), \quad \boldsymbol{k}_3 = (-1, -3), \\ &\boldsymbol{Y}_0 + \tfrac{1}{2}\boldsymbol{k}_3 = (\tfrac{1}{2}, -\tfrac{3}{2}), \quad \boldsymbol{k}_4 = (-\tfrac{3}{2}, -2). \end{aligned}$$

Then $\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4 = (0 - 2 - 2 - \frac{3}{2},\ -4 - 8 - 6 - 2) = (-\frac{11}{2}, -20)$ and

$$\boldsymbol{Y}_1 = (1, 0) + \tfrac{1}{12}(-\tfrac{11}{2}, -20) = \Big(\frac{13}{24}, -\frac{5}{3}\Big) = (0.5417, -1.6667).$$

Distances from the true point $(0.5403, -1.6829)$: RK4 0.016, trapezoidal 0.10, forward Euler 0.56, backward Euler 0.68; leapfrog's $U_1 = 0.5$ is 0.040 from $\cos 1$ (its velocity sits at a different time). RK4 lands closest, by a wide margin, at the cost of four evaluations per step. (Its eigenvalues have $|\lambda| = 0.9939$ and $\theta = 56.98°$, so even at $\omega h = 1$ it loses 0.6% of the amplitude and 0.3° per step.)

Question 4 Energy in phase space

(a) $\dot{u} = v$, $\dot{v} = -u$:

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad E(0) = 1 \text{ J}.$$

The true solution $u = \cos t$, $v = -\sin t$ keeps $E = 1$, so the orbit is the unit circle $u^2 + v^2 = 1$, through $u = \pm 1$ and $v = \pm 1$. At $(1, 0)$ the velocity $\dot{v} = -1 < 0$ points down, so the motion is clockwise. The period is $T = 2\pi/\omega = 2\pi \approx 6.28$ s. (With $m = c$ the energy is the squared radius, so this is the circle of the lectures.)

(b) $\boldsymbol{G}_F = \boldsymbol{I} + h\boldsymbol{A} = \left[\begin{smallmatrix} 1 & 0.25 \\ -0.25 & 1 \end{smallmatrix}\right]$.

$$\begin{aligned} &\boldsymbol{Y}_1 = (1, -0.25), && E_1 = 1 + 0.0625 = 1.0625, \\ &\boldsymbol{Y}_2 = (1 - 0.0625,\ -0.25 - 0.25) = (0.9375, -0.5), && E_2 = 0.87890625 + 0.25 = 1.12890625. \end{aligned}$$

$E_1/E_0 = 1.0625$ and $E_2/E_1 = 1.12890625/1.0625 = 1.0625$: the energy grows by the factor $1 + \omega^2 h^2 = 1 + h^2 = 1.0625$ at every step. (In general $(v - hu)^2 + (u + hv)^2 = (1 + h^2)(u^2 + v^2)$, the cross terms $-2huv + 2huv$ cancelling.) It doubles when $1.0625^n = 2$, that is $n = \ln 2 / \ln 1.0625 = 11.4$, so by the 12th step (at $t = 3$ s, about half a period). One period is $2\pi/0.25 = 25.1$ steps, so E grows by $1.0625^{25.1} \approx 4.6$ per period (the amplitude by $\sqrt{4.6} \approx 2.1$).

(c) $\boldsymbol{I} - h\boldsymbol{A} = \left[\begin{smallmatrix} 1 & -0.25 \\ 0.25 & 1 \end{smallmatrix}\right]$ has determinant 1.0625, so

$$\boldsymbol{G}_B = \frac{1}{1.0625}\begin{bmatrix} 1 & 0.25 \\ -0.25 & 1 \end{bmatrix} = \frac{\boldsymbol{G}_F}{1.0625}, \qquad \boldsymbol{Y}_1 = \frac{(1, -0.25)}{1.0625} = (0.9412, -0.2353),$$

and $E_1 = 1.0625/1.0625^2 = 1/1.0625 = 0.9412$. Backward Euler divides the energy by 1.0625 at every step (down to 0.22 of its value after one period).

Trapezoidal with $h = 0.5$: $\frac{h}{2}\boldsymbol{A} = \left[\begin{smallmatrix} 0 & 0.25 \\ -0.25 & 0 \end{smallmatrix}\right]$, and $\boldsymbol{I} - \frac{h}{2}\boldsymbol{A} = \left[\begin{smallmatrix} 1 & -0.25 \\ 0.25 & 1 \end{smallmatrix}\right]$ has determinant $1.0625 = 17/16$, so

$$\boldsymbol{G}_T = \frac{16}{17}\begin{bmatrix} 1 & 0.25 \\ -0.25 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0.25 \\ -0.25 & 1 \end{bmatrix} = \frac{16}{17}\begin{bmatrix} 0.9375 & 0.5 \\ -0.5 & 0.9375 \end{bmatrix} = \frac{1}{17}\begin{bmatrix} 15 & 8 \\ -8 & 15 \end{bmatrix},$$

and $\boldsymbol{Y}_1 = (\frac{15}{17}, -\frac{8}{17}) = (0.882, -0.471)$. Check: $u^2 + v^2 = \frac{225+64}{289} = 1$, so $E_1 = 1$. The point is on the circle: the trapezoidal method keeps the energy exactly. (The true point at $t = 0.5$ is $(\cos 0.5, -\sin 0.5) = (0.878, -0.479)$, a little further round; the error of the trapezoidal method is in the angle, not in the energy.)

(d) With $h = 0.2$:

$$V_{1/2} = 0 - 0.1 \cdot 1 = -0.1, \qquad U_1 = 1 + 0.2(-0.1) = 0.98,$$
$$V_{3/2} = -0.1 - 0.2 \cdot 0.98 = -0.296, \quad U_2 = 0.98 + 0.2(-0.296) = 0.9208.$$

$$Q_0 = (1)(0.98) + (0.1)^2 = 0.98 + 0.01 = 0.99,$$
$$Q_1 = (0.98)(0.9208) + (0.296)^2 = 0.902384 + 0.087616 = 0.99.$$

The two agree: Q is the quantity leapfrog conserves exactly, at every step. It is the energy $E = u^2+v^2$ with u^2 replaced by the product of two neighbouring positions, and $Q_0 = E_0(1-h^2/4) = 0.99$. Leapfrog is stable for $\omega h < 2$, that is $h < 2$ s; the period is 2π s, so that means more than π, hence at least 4, steps per period.

Over many periods: forward Euler's E grows without bound (times 1.0625 per step here); backward Euler's decays to zero (divided by 1.0625 per step); leapfrog's stays bounded, moving up and down within a fixed band around E_0 while Q is exactly constant; the trapezoidal method's is exactly constant, $E = 1$ at every step and for every h.